

Thesis

Sprint 2 - Experiment Results & Avenues

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The Game

During experiments

- 30 LLM agents on a spatial grid
- Each round they choose ONE action. Simultaneous execution.
- Goal: Optimize Resources
- 50 rounds, known endpoint
- Neutral prompt, identical for all
- Agents see own resources, neighbours resources, neighbours recent actions
- **Invest self is turned off. Leads to stalemate (only invest self, gini = 0.000).**

Action	Cost	Effect	Risk
Invest_other	0	+5 to target	None
arm_self	2	2x defense/ offense, 3 rounds	Opportunity Cost
Attack	0	Take 40% of target's resources	Might lose 40%
Arm_other	2	Arm an ally, 2x extra	Might use to attack you
Do Nothing	0	Do nothing	Fall behind

Phase 1: Characterisation

142 OpenRouter runs, 6 parameter sweeps, 3 models

- Out of the parameter sweeps, 2 clear phase transitions are found (affecting gini). Other params had no clear effect (on Nova Micro, arguably too dumb to understand structure)
- Alternating the models (Gemini Flash, Nova Micro) yielded substantial effects (Gemini cooperates 10x more).
- Too many confounds in comparing models, so decided thats out of scope. Transitioned to Gemma 2 27B on Snellius.
- Adding spatial effects (random walk for agents) had significant effects
- 7x7 grid (full observability = R3+), 10 agents

Parameter	Transition	Effect
Attack take %	T=10 -> T=20	Gini 0.20 -> 0.46
Invest Self	ON -> OFF	Stalemate -> rich dynamics

Radius	Coop%	Attack%	Gini
R=1 (~20% visible)	52	6	.25
R=2 (~50% visible)	60	13	.48
R=3 (100%)	20	55	.71

Phase 2: Framing

100 Gemma 2 27B runs, 5 framings × 20 reps, single model

- Framing explains 90% of gini variance ($\eta^2 = 0.901$)
- System is **very sensitive** to prompting, as expected

Framing	Gini (mean±std)	Coop%	Dominant Behaviour
cooperative	0.581 ± 0.034	89%	invest_other (59%)
cautious	0.549 ± 0.053	79%	invest_other + do_nothing
neutral	0.778 ± 0.033	68%	invest_other (56%)
strategic	0.725 ± 0.034	50%	invest_other (38%) + arm (28%)
competitive	0.858 ± 0.037	37%	invest (32%) + attack (30%)

Phase 3a: Reasoning Depth Pilot (old prompting)

12 runs, 4 levels × 3 reps, Gemma 2 27B, arm_cost=0

- More reasoning is more conflict, but non-monotonic. L2 has less attacking than L1 and L3, but more arming!
- Fear at deeper reasoning levels (L2)
 - “Several nearby agents are armed and likely to attack. I will arm myself.”
 - "Many agents are arming themselves. I will arm myself to be prepared.”
 - "Agent 1 is likely to be targeted. I will arm to deter attacks."
- Expressed through arming instead of pre-emptive striking (perhaps 8% in L3 are pre-emptive strikes?)
- Hypothesis: without arm, L2 would also have pre-emptive strikes

Level	Instruction (core)	Gini	Coop %	Arm %	Atk%
L0	State choice briefly, don't deliberate	.566	70%	0%	5%
L1	Calculate expected value per action	.632	63%	4%	6%
L2	Predict what neighbors will do	.624	39%	40%	2%
L3	They're also reasoning about you	.712	52%	16%	8%

Phase 3b: Reasoning × Game Structure

132 runs, 4 parameter sweeps × 4 reasoning levels, NEW PROMPT

- Deeper reasoning seems to amplify structure incentive
- L2 is very sensitive to increased arm cost.
-> arming is substituted to other actions
- Investing still has strongest effect on outcomes.
- In ‘safe’ regime, L3 still stays exploitative where L0 doesn’t!

Level	arm_cost=0	arm_cost=5	Δ Gini
L0	.606	.561	-.045
L1	.616	.702	+.086
L2	.578	.804	+.226
L3	.729	.785	+.056

Level	invest_self OFF	invest_self ON	Δ Gini
L0	.595	.184	-.411
L3	.703	.526	-.177

Old vs. New Prompts

12 runs, 4 levels × 3 reps, new prompt structure + arm_cost=2, Confound: two things changed

- Old prompt structure used raw data dump as history. Agents reasoning at L2 and L3 was not super coherent. Changed the prompt to have a structured history of themselves and neighbours. Cut the context by 60%. Reasoning looks more logical.
- Just found out I created a confound in comparison: arming + new prompt.

Old prompt - (arm_cost = 0)

Level	Gini	Coop%	Arm%	Atk%
L0	.566	70%	0%	5%
L1	.632	63%	4%	6%
L2	.624	39%	40%	2%
L3	.712	52%	16%	8%

New prompt - (arm_cost = 2)

Level	Gini	Coop%	Arm%	Atk%
L0	.712	72%	13%	7%
L1	.740	53%	17%	14%
L2	.615	68%	20%	3%
L3	.574	64%	20%	6%

- Shifted patterns: L0/L1 are now most aggressive. L1 produces most inequality. Arming is done by all levels now.
- L2 hoarder is now gone (40% -> 20% arming, but could be the confounding variable)

Pre-emptive attacks and action stability

Pre-emptive: attacking someone that hasn't attacked you before

Pre-emptive striking

- Almost all attacks are pre-emptive -> makes sense with 30 agents, rarely see the same ones?
- L2 least attacks but most pre-emptive. Not sure if that says anything though.

Level	Total Atks	Pre-emptive%	1st Atk Round
L0	109	93%	2.0
L1	207	83%	1.0
L2	41	97%	3.3
L3	84	87%	2.0

Action Stability

- Higher level means less stability -> more differing actions.
- Agents don't stabilise in their actions.

Level	Start (r2-11)	End (r41-50)	Trend
L0	82%	67%	↓ -15%
L1	71%	58%	↓ -13%
L2	65%	49%	↓ -17%
L3	55%	45%	↓ -10%

Where we stand

What we know

- It shows the system creates rich dynamics. Found some interesting phase transitions, especially with radius.
- Framing explains 90% of the variance. Reasoning depth through prompting changes behaviour.
- Deeper reasoning amplifies structure.

Whats uncertain

- Reasoning depth is essentially prompting. This is hard to generalise.
- We saw different models produce wildly different outcomes. Any outcome is hard to generalise.
- Faithfulness of CoT is still unvalidated
- Do I want to do a context engineering thesis?

Done

- 400+ runs
- Game engine, spatial grid, reasoning levels, metrics pipeline.
- Parameter sensitivity mapped
- Good Gemma 2 27B fingerprint
- Switch to Qwen 3.5 35B! Just released, and apparently very good.

Behind

- Literature
- Nothing written yet for thesis
- Direction! What do I investigate!!?

Two directions

Stay in prompt engineering or move to structural effects?

Reasoning Depth x Structure

Design: 4 levels x 6 radii x 20 reps

Clean design, results already partially collected.

Results tied to prompt engineering.
Probably not robust. K-level via instruction is not real reasoning.

Credible Commitment

Design: 6 collateral levels x 20 reps

- Prompting becomes less relevant.
- Agent architecture/design identical. Only environment changes.
- Possibly generalisable. Relevant across models.

Credible Commitment

Same game, same agents — add one mechanism: enforceable contracts.
Why: framing is responsible for variation, reasoning level does not create stable cooperation, but deepens structural incentives. Will institutional contracts solve trust issues?

- Agents do not converge to cooperation -> either arming, or pre-emptively attacking
- Credible commitment could solve a trust issue, which might be relevant for future autonomous AI agent interaction
- Maybe enforceable contracts make agents escape Hobbesian trap through structure, not prompting
- Can structure provide stable cooperation in asymmetric information environments?

Contract mechanics

1. Agent proposes contract to neighbor ('we invest in each other this round')
2. Target accepts or refuses
3. If accepted: collateral locked
4. Deviation = lose collateral
5. One contract per agent per round
6. Contract lasts one round

Mechanism research could include investigating effects of:

- multi-party contracts
- Assymetrical contracts (I protect, you pay)
- Contract duration

Condition	Description	Prediction
No contract	Baseline anarchy	War equilibrium
0% collateral	Cheap talk (promise, no enforcement)	Minimal effect
10% collateral	Light commitment	Some cooperation
25% collateral	Moderate commitment	Transition zone?
50% collateral	Serious commitment	Stable cooperation ?
100% collateral	Full commitment	Maximum cooperation

Questions

- Which direction do you prefer?
- If A: is instructed K-level enough/defensible?
- If B: is collateral % a clean enough manipulation, or should we do something else? We could make Credible Commitment the core and investigate adding mechanisms.
- How to choose base params? Most cooperation, most war?
- Scope: is one factorial enough for the thesis?